

Quiz navigation

1 2 3 4

Show one page at a time

Finish review

Course Home

Course Info ▲

Syllabus

Participants list

Grade/Attendance ▲

Offline-Attendance

Grades

Students Notifications ▼

Others ▼

Activities/Resources +



Linear Algebra

3Week [18 September - 24 September]

Quiz 3

Started on	2023-09-25 00:44
State	Finished
Completed on	2023-09-25 01:02
Time taken	18 mins 10 secs
Marks	2.33/4.00
Grade	1.17 out of 2.00 (58%)

Question 1

Partially correct

Mark 0.67 out of 1.00

Let A and B be $n \times n$ matrices. I is identity matrix.

Find true statements.

Select one or more:

- ☒ $(A^T)^T = A$ ✓
- ☐ If $A \cdot B = 0$ then either $A = 0$ or $B = 0$.
- ☒ $(A + B)^T = B^T + A^T$ ✓
- ☐ $(A + I)^2 = A^2 + 2A + I$
- ☐ $(A + B)^2 = A^2 + 2AB + B^2$

Your answer is partially correct.

You have correctly selected 2.

The correct answers are: $(A^T)^T = A$,

$(A + B)^T = B^T + A^T$, $(A + I)^2 = A^2 + 2A + I$

Question 2

Correct

Mark 1.00 out of 1.00

Find $3A - 4B$, if $A = \begin{bmatrix} 1 & 3 \\ a & b \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ -3 & 0 \end{bmatrix}$.

Select one:

- ☒ $\begin{bmatrix} -1 & -7 \\ 3a + 12 & 3b \end{bmatrix}$ ✓
- ☐ impossible
- ☐ $\begin{bmatrix} -1 & 7 \\ 3a - 12 & 3b \end{bmatrix}$

Your answer is correct.

The correct answer is: $\begin{bmatrix} -1 & -7 \\ 3a + 12 & 3b \end{bmatrix}$

Question 3
Partially correct
Mark 0.67 out of 1.00

Find true statements, if A is 2×3 matrix and B is 3×4 matrix.

Select one or more:

- ☐ Result of their multiplication is 3×3 matrix
- ☒ Multiplication is possible ✓
- ☐ Addition is possible
- ☒ Product is 2×4 matrix ✓
- ☐ Subtraction is impossible

Your answer is partially correct.

You have correctly selected 2.

The correct answers are: Subtraction is impossible, Multiplication is possible, Product is 2×4 matrix

Question 4
Incorrect
Mark 0.00 out of 1.00

Let A and B be $m \times n$ matrices. Find true statements.

Select one or more:

- ☒ $(A^{-1})^{-1} = A$ ✗
- ☐ $(A \cdot B)^{-1} = A^{-1} \cdot B^{-1}$
- ☒ $(A^T)^{-1} = (A^{-1})^T$ ✗
- ☐ none

Your answer is incorrect.

The correct answer is: none